

# Watson Hartsoe

## Research Engineer and Creative Technologist

Atlanta, GA   watson.hartsoe@gmail.com   Portfolio   GitHub   LinkedIn

I build browser tools and Python pipelines for generative media, interactive 3D, and film production. My work combines software development, privacy and participatory research, and coordination of creative and technical teams. PhD candidate in Digital Media at Georgia Tech.

### RESEARCH AND PROFESSIONAL EXPERIENCE

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#### Georgia Institute of Technology

2023–present

Doctoral Researcher, Digital Media · Atlanta, GA

- Develop thick prompting, a research method for studying generative workflows through documented prompt variation, output comparison, and traceable media records.
- Coauthored participatory Emotion AI auditing research published at NordiCHI 2024; develop interactive privacy and ethics prototypes.
- Instructor, Intro to Media Studies (LMC 2400 B1), Spring 2026; mentor undergraduate researchers in the Art & AI VIP Program.

#### Xsolla

Jan–Jun 2023

Consultant · Babka UX Researcher, Trust, Privacy & Security; Nushi AI Artist

- Built and tested a system of 200+ custom chatbots through company-wide workshops exploring privacy in conversational commerce.
- Created AI image transformations for Nushi at GDC 2023, supporting camera-based visuals, on-site displays, and livestreams; built Python/MoviePy video preparation tools for ControlNet and Runway GEN-1.
- Served as a deputized producer, coordinating photographers, roboticists, performers, and media handoffs across capture, AI processing, and delivery. Campaign reach reported by the creative producer: 55M+ views and 465K+ followers in 10 days.

#### Genesis STEAM

2016–2021

Lead Instructor · Los Angeles, CA

- Created curriculum and training resources for teachers and nonprofits; taught prototyping, computational thinking, 360 filmmaking, and virtual placemaking with museum, design, and development teams.
- Co-inventor, published U.S. patent application US20210241650A1 (2021): Collaborative system for teaching design and critical thinking skills.

#### Ojo Joven and Artes Sin Fronteras

2013–2015

Founder, Ojo Joven · Fulbright Scholar; Princeton in Latin America Fellow · Honduras

- Founded and managed a participatory youth photography program; developed curriculum and media methods, led training, and facilitated community conversations.
- Conducted year-long ethnographic fieldwork using participant observation, focus groups, and interviews. Presented community media methods at the Community Development Society Conference (2015).

Additional experience: Emory research collaboration (2023–2024); IDDS sanitation research and co-design, Guatemala (2017); Episcopal School of Los Angeles Teaching Fellow (2015–2016).

### EDUCATION

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PhD candidate, Digital Media

Georgia Institute of Technology · 2023–present

SELECTED RESEARCH AND TECHNICAL WORK

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CINEOSIS and *Where You Go When You Leave* Demo · Code

Chief AI Operations with Mark Anthony Thomas; feature film in production. Built browser tools for searching, comparing, and assembling footage with synchronized playback, editable branches, source tracking, and EDL export; Python media workflows use OpenCLIP, SAM 2.1, OpenCV, and FFmpeg.

WAG and LDraw Authoring Tools Editor · Code

Built JavaScript/Three.js interfaces linking editable LDraw source with 3D scenes: part selection, protected edits, undo/redo, and independent scene states.

Privacy Nutrition Label Source

Built a browser prototype translating contextual and multidimensional privacy concepts into editable labels, comparison views, and learning scenarios, with JSON import/export. Richmond Wong’s Project Studio, 2025.

LEGOS Project · Code

Developed a narrative grammar and browser tools for specifying locations, entities, goals, obstacles, shifts, and solutions. Presented in the TRANSFORMER 2025 workshop; a precursor to related ELO 2026 work.

SELECTED PUBLICATIONS, PRESENTATIONS AND RECOGNITION

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- Reflective Design for Informal Participatory Algorithm Auditing: A Case Study with Emotion AI. With Noura Howell, Jacob Amin, and Vyshnavi Namani. NordiCHI 2024; published paper.
- Coaxing the Ripples: thoughts on distant writing of interactive works. With Terhi Marttila and Jay David Bolter. ELO, July 2026; conference talk.
- Forging Realities: AI as Material, Memory, and Resistance. TRANSFORMER, 2025; workshop leader.
- The AI Box Experiment: Centaurs and Synthetic Humans. Georgia Tech Digital Media, February 2024; talk.
- STEM Pop-Up Lab. With Magalis Martinez. Talks at Google, 2016.
- ImmerseGT winner, 2023: WAGYO. Co-created with Yago Arconada; AI-assisted 3D scene creation.

SKILLS

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| Programming and graphics   | JavaScript, Python, HTML/CSS, Three.js, WebGL, GLSL, Canvas, SVG, Web Audio.                                                                            |
| AI and media               | LLM APIs, prompt/context engineering, OpenCLIP, SAM 2.1, OpenCV, FFmpeg, MoviePy, ControlNet.                                                           |
| Interfaces and delivery    | Responsive interfaces, WebSockets, WebRTC, Git/GitHub, GitHub Pages.                                                                                    |
| Research and collaboration | Ethnography, interviews, focus groups, usability testing, participatory design, qualitative synthesis, technical facilitation, production coordination. |
| Languages                  | English; Spanish, high proficiency.                                                                                                                     |